Beyond T90: Using the Bp–P0 Plane to Diagnose Anomalous GRB Central Engines

Steve Girard

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Abstract

The traditional short/long classification of gamma-ray bursts (GRBs), based primarily on T_{90}, is increasingly insufficient for interpreting anomalous events that combine long-duration prompt emission with merger-like signatures, extended central-engine activity, kilonova-like components, atypical host environments, or the absence of an accompanying supernova. Recent work on magnetar-powered GRBs has introduced an empirical B_p–P_0 relation for short GRBs with internal X-ray plateaus, showing that short and long GRBs may share a similar scaling slope while occupying different magnetic-field normalizations. This paper proposes the B_p–P_0 plane as an additional central-engine diagnostic for anomalous GRBs. Using GRB 211211A and GRB 250702B as motivating cases, and GRB 241025A as a methodological reminder of model non-specificity, we argue that ambiguous GRBs should be evaluated not only by duration and environmental diagnostics, but also by the inferred or expected location of their engine in magnetar parameter space whenever plateau or late-injection observables permit such a test.

Keywords: gamma-ray bursts; magnetars; central engines; T_{90}; short GRBs; long GRBs; kilonovae; anomalous transients; B_p–P_0 relation; model degeneracy.

1 Introduction

1.1 The limits of the T_{90} classification

Gamma-ray bursts are traditionally divided into short and long classes according to their duration, commonly measured by T_{90}. In the standard interpretation, short GRBs are associated primarily with compact-object mergers, while long GRBs are associated with the collapse of massive stars. This classification remains operationally useful, but recent events show that duration alone is not a sufficient physical diagnostic.

Several anomalous GRBs combine observables that cut across the canonical short/long division. GRB 211211A, for example, displayed a long prompt duration while also showing evidence consistent with a compact-merger origin, including a kilonova-like component, no accompanying supernova,

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and prolonged central-engine activity (Troja et al., 2022; Gompertz et al., 2022; Hamidani et al., 2024; Kunert et al., 2024; Waxman et al., 2025; Girard, 2026a). GRB 250702B presents a different anomaly: an ultra-long, multi-episode burst with quasi-regular pulse separation, extreme obscuration, and a disturbed host environment, for which a single magnetar engine embedded in a non-axisymmetric environment may offer a parsimonious interpretation (Girard, 2026b; Granot et al., 2026).

These cases illustrate a broader problem: phenomenological classification by duration, host environment, supernova or kilonova association, and afterglow morphology may remain underdetermined when several physical narratives can reproduce the same observational pattern. A further diagnostic axis is therefore needed, one that probes the central engine more directly.

1.2 A central-engine diagnostic beyond duration

Recent work on magnetar-powered GRBs provides such a possible diagnostic. In particular, the inferred relation between the polar magnetic field B_p and the initial spin period P_0 of newborn magnetars offers a quantitative engine-parameter space. For short GRBs with internal X-ray plateaus, Li et al. report a B_p–P_0 correlation whose slope is nearly identical to that of long GRBs, but whose normalization is shifted toward substantially larger magnetic fields (Li et al., 2026).

This result suggests that short and long GRBs may share a broadly similar spin-down mechanism while differing in progenitor channel, accretion environment, or magnetic-field normalization. The B_p–P_0 plane may therefore function not merely as a magnetar compatibility check, but as a diagnostic space in which anomalous GRBs can be compared to merger-like, collapsar-like, or intermediate engine populations.

1.3 Aim of this paper

This paper proposes the B_p–P_0 plane as a central-engine diagnostic for anomalous GRBs whose phenomenology crosses the conventional short/long boundary. The goal is not to replace existing diagnostics, nor to introduce a new rigid classification scheme. Rather, the goal is to provide a bridge between phenomenological classification and engine-level parameter inference.

The working claim is simple:

> Anomalous GRBs should be evaluated not only by T_{90}, host properties, kilonova or supernova associations, and afterglow morphology, but also by their inferred or expected location in the magnetar B_p–P_0 plane whenever plateau or late-injection observables permit such an inference.

GRB 211211A and GRB 250702B are used as motivating cases. GRB 241025A is used more narrowly, as a methodological reminder that a single multi-wavelength dataset can support multiple coherent theoretical narratives, thereby motivating the search for specificity-enhancing diagnostics.

2 The B_p–P_0 Plane as a Magnetar-Engine Diagnostic

2.1 Basic magnetar spin-down relations

In the standard magnetar spin-down interpretation of internal X-ray plateaus, the characteristic spin-down timescale τ and spin-down luminosity L_0 are related to the magnetar's initial angular velocity Ω_0, initial spin period P_0, and polar magnetic field B_p (Lyons et al., 2010; Rowlinson et al., 2014; Lü et al., 2018; Li et al., 2026).

The characteristic spin-down timescale may be written as

$$\tau = \frac{3c^3 I}{B_p^2 R^6 \Omega_0^2} = \frac{3c^3 I P_0^2}{4\pi^2 B_p^2 R^6}, \tag{1}$$

or, in normalized form,

$$\tau \simeq 2.05 \times 10^3 \text{ s}\, I_{45} B_{p,15}^{-2} P_{0,-3}^2 R_6^{-6}. \tag{2}$$

The characteristic spin-down luminosity is

$$L_0 = \frac{I\Omega_0^2}{2\tau} \simeq 1.0 \times 10^{49} \text{ erg s}^{-1}\, B_{p,15}^2 P_{0,-3}^{-4} R_6^6. \tag{3}$$

For X-ray plateau applications, one typically identifies

$$\tau = \frac{t_b}{1+z}, \tag{4}$$

where t_b is the observed plateau break time and z is the redshift, and approximates

$$L_0 \simeq L_b, \tag{5}$$

where L_b is the plateau luminosity at the break.

The inferred magnetar parameters may then be written schematically as

$$B_{p,15} \simeq 2.05\, I_{45} R_6^{-3} L_{0,49}^{-1/2} \tau_3^{-1}, \tag{6}$$

and

$$P_{0,-3} \simeq 1.42\, I_{45}^{1/2} L_{0,49}^{-1/2} \tau_3^{-1/2}. \tag{7}$$

These relations turn plateau observables into inferred engine parameters. The diagnostic power of the method therefore depends on the quality of the plateau identification, the redshift, the luminosity estimate, and the validity of the magnetar spin-down interpretation.

2.2 Empirical B_p–P_0 relations for short and long GRBs

Li et al. report that short GRB magnetars occupy the following approximate parameter ranges (Li et al., 2026):

$$P_0 \in [1.73, 18.28] \text{ ms}, \tag{8}$$

and

$$B_p \in [0.06, 2.82] \times 10^{17} \text{ G}, \tag{9}$$

with an average magnetic field

$$\langle B_p \rangle \simeq 7.05 \times 10^{16} \text{ G}. \tag{10}$$

For long GRBs, the comparison sample occupies approximately

$$P_0 \in [0.95, 13.79] \text{ ms}, \tag{11}$$

and

$$B_p \in [0.39, 23.08] \times 10^{15} \text{ G}, \tag{12}$$

with an average magnetic field

$$\langle B_p \rangle \simeq 3.69 \times 10^{15} \text{ G}. \tag{13}$$

The fitted short-GRB relation is (Li et al., 2026)

$$\log B_p = (0.84 \pm 0.07) \log P_0 + (15.79 \pm 0.07), \tag{14}$$

whereas the corresponding long-GRB relation is

$$\log B_p = (0.83 \pm 0.09) \log P_0 + (14.92 \pm 0.06). \tag{15}$$

The slopes are nearly identical, while the intercepts differ. This implies that the two GRB populations may share a common spin-down scaling while occupying different magnetic-field normalizations. Because both B_p and P_0 are inferred from plateau observables under a specific spin-down model, this separation should be treated as a diagnostic hypothesis rather than as a model-independent classification boundary.

Figure 1 summarizes this diagnostic separation between the short-GRB and long-GRB magnetar loci.

2.3 Diagnostic interpretation

The central diagnostic value of the B_p–P_0 plane lies in the distinction between scaling and normalization. If the common slope reflects a broadly universal magnetar spin-down mechanism, then the vertical offset may encode differences in progenitor channel, accretion rate, magnetic-field amplification, or compact-object formation environment.

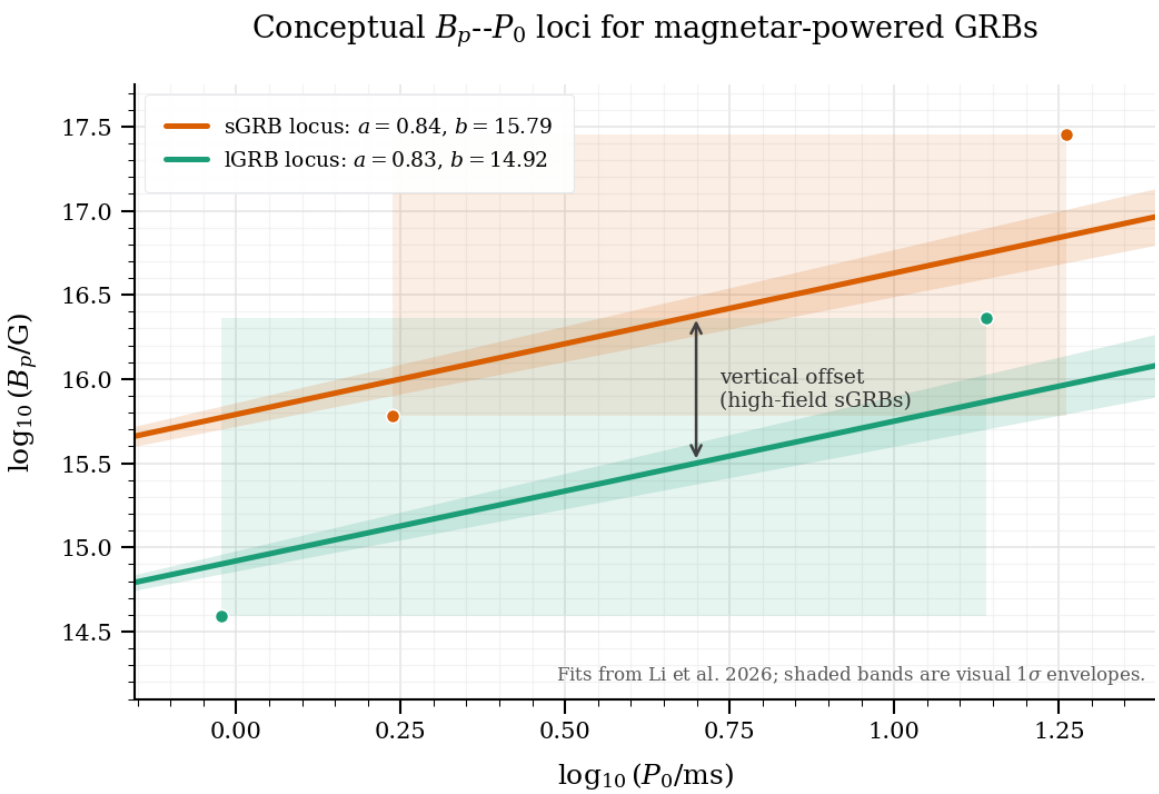

Figure 1: Conceptual B_p–P_0 loci for magnetar-powered GRBs. The short-GRB and long-GRB relations have nearly identical slopes, but the short-GRB locus is shifted toward higher magnetic-field normalization. Fits and quoted parameter ranges are based on Li et al. (2026).

Thus, for anomalous events, the relevant question is not merely whether a magnetar engine is possible. The more specific question is:

> If a magnetar engine is invoked, does the inferred engine resemble the short-GRB high-field population, the long-GRB lower-field population, an intermediate regime, or an outlier?

This reframes the B_p–P_0 plane as an engine-space diagnostic rather than a simple phenomenological correlation.

3 Why Anomalous GRBs Need an Engine-Space Test

3.1 Classical diagnostics and their failure modes

Anomalous GRBs are usually evaluated through a combination of diagnostics. Each diagnostic carries useful information, but each also has a characteristic failure mode. The difficulty is not that these criteria are useless, but that none of them is individually decisive when the event crosses the conventional short/long boundary.

- **Duration, especially T_{90}.** Duration provides the first and simplest classification gate. In the canonical picture, short durations suggest compact-object mergers, whereas long durations suggest massive-star collapse. However, events such as GRB 211211A show that duration can be misleading when a long prompt phase is accompanied by merger-like signatures.
- **Prompt spectral hardness and spectral lag.** Prompt spectral properties can provide additional clues about the physical class of the burst. Short-like hardness or near-zero spectral lag may support a merger-like interpretation even when the observed duration is long. Yet these quantities are not sufficient by themselves, because prompt-emission properties can overlap across populations.
- **Extended emission.** Extended emission can indicate prolonged activity beyond the initial prompt spike and may point toward a long-lived central engine. However, extended emission does not uniquely determine the nature of that engine. It can be interpreted through magnetar spin-down, fallback accretion, or other late-time energy-injection mechanisms.
- **X-ray plateaus or steep post-plateau declines.** X-ray plateaus are especially important because they may encode the spin-down timescale and luminosity of a newborn magnetar. A steep decay following the plateau can strengthen the case for an internal origin. The limitation is that not every plateau is necessarily magnetar-powered or cleanly separable from the external afterglow.
- **Kilonova or supernova association.** A kilonova component strongly favours compact-merger ejecta, while a supernova association supports a massive-star collapse origin. These diagnostics are powerful because they probe the progenitor channel directly. Their failure mode is observational: such components can be faint, obscured, missed because of cadence or wavelength coverage, or hidden by host-galaxy contamination.

- **Host-galaxy type, star-formation rate, and morphology.** The host environment provides population-level context. A star-forming host may be more naturally associated with collapsars, while low star formation, disturbed morphology, or an older stellar population may support merger-related scenarios. However, host properties are indirect diagnostics and cannot determine the central engine on their own.

- **Galactocentric offset.** A large offset from the host-galaxy centre can support a compact-binary origin, since binary systems may travel far from their birth sites before merging. Conversely, a small offset may be more consistent with massive-star collapse. The limitation is that offsets are probabilistic rather than definitive, and projection effects or host complexity can weaken the inference.

- **Afterglow behaviour across X-ray, optical, near-infrared, radio, and GeV bands.** Multi-wavelength afterglow modelling can constrain the circumburst medium, jet geometry, microphysical parameters, and possible late energy injection (Sari et al., 1998). It is one of the richest diagnostic tools available. However, afterglow fits often contain degeneracies: different combinations of energy, density, microphysics, viewing geometry, and additional components can reproduce similar light curves.

- **Evidence for late energy injection or rebrightening.** Late-time flattening, rebrightening, or high-energy activity can suggest continued central-engine power. Such behaviour is particularly relevant for testing magnetar or fallback scenarios. The difficulty is that late emission can also arise from density structure, refreshed shocks, jet effects, or multiple emitting components.

Taken together, these diagnostics can strongly constrain the interpretation of an anomalous GRB. Yet their evidential force remains distributed across phenomenology, environment, and afterglow modelling. This is why an additional engine-level diagnostic is useful: the B_p–P_0 plane does not replace these criteria, but tests whether a proposed magnetar engine occupies a plausible and population-consistent region of parameter space.

3.2 Model non-specificity

A methodological problem arises when several independently motivated models can reproduce the same target pattern. GRB 241025A provides an instructive example. Its radio colour evolution, optical and near-infrared light curves, X-ray behaviour, and broadband SEDs can be interpreted within a standard synchrotron external-shock framework, but a more complex Binary-driven HyperNova interpretation may also remain coherent under different assumptions (Giarratana et al., 2026; Ruffini et al., 2024; Girard and Kline, 2026).

The point is not that model multiplicity is a failure. Rather, it shows that phenomenological classification may remain underdetermined when competing narratives reproduce the same data. In such cases, additional constraints are needed to reduce interpretive non-specificity.

3.3 The B_p–P_0 plane as an additional constraint

The B_p–P_0 plane provides one such constraint when a magnetar interpretation is observationally plausible. It does not replace host, kilonova, supernova, or afterglow diagnostics. Instead, it asks whether the central engine required by the model occupies a physically plausible and population-consistent region of magnetar parameter space. The possible outcomes are:

- **Short-GRB locus.** If an anomalous event falls near the short-GRB locus, its inferred engine parameters may be compatible with the high-field magnetars associated with compact-merger systems. This would not prove a merger origin by itself, but it would strengthen the case that the event is engine-compatible with the short-GRB population rather than with ordinary collapsar-like long GRBs.

- **Long-GRB locus.** If the inferred parameters fall near the long-GRB locus, the event may be more naturally compatible with a collapsar-like magnetar engine. This outcome would support the view that the event is anomalous primarily in its observed phenomenology, environment, or afterglow behaviour, rather than in its underlying engine class.

- **Intermediate region.** An intermediate location between the short- and long-GRB loci would be especially informative. It could indicate a transitional or hybrid engine regime, or it may suggest that the event combines features of different progenitor channels. Such a result would not define a new class by itself, but it would identify the event as a priority case for further modelling and follow-up.

- **Outlier regime.** If the inferred parameters fall well outside both loci, the magnetar interpretation would require additional scrutiny. The event might demand extreme magnetic fields, unusual initial spin periods, non-standard assumptions about the neutron star, or an alternative central-engine model. In this case, the B_p–P_0 plane would act less as a classifier than as a stress test of the proposed interpretation.

The diagnostic is therefore best understood as a consistency and specificity test, not as a standalone progenitor classifier. Its value lies in asking whether a proposed magnetar engine is merely qualitatively plausible, or whether it occupies a region of parameter space consistent with known GRB engine populations.

4 Motivating Case I: GRB 211211A as a Long-Duration Merger-like Event

4.1 Why GRB 211211A is relevant

GRB 211211A illustrates the failure of a duration-only classification (Troja et al., 2022; Gompertz et al., 2022; Girard, 2026a). Although the event had a long prompt duration, its multi-wavelength properties favour a compact-merger interpretation. Its relevance lies in the fact that several independent diagnostics point away from a simple collapsar-like reading despite the long duration. The key features include:

- **Long prompt emission with extended activity.** The long duration initially places GRB 211211A on the long-GRB side of the conventional classification. However, the presence of extended activity complicates this interpretation, because prolonged emission can arise from a central engine whose lifetime is not simply tied to the prompt duration class.
- **Short-like prompt properties despite long duration.** GRB 211211A combines a long observed duration with prompt-emission properties more commonly associated with short GRBs. This mismatch is precisely what makes the event diagnostically important: it shows that duration and prompt physical character need not point to the same progenitor class.
- **A near-infrared excess consistent with a kilonova component.** A kilonova-like near-infrared excess provides one of the strongest indications of compact-merger ejecta (Troja et al., 2022; Gompertz et al., 2022; Hamidani et al., 2024; Waxman et al., 2025). If this component is correctly identified, it shifts the interpretation away from an ordinary massive-star collapse event, even though the burst duration is long.
- **Absence of an accompanying supernova.** The lack of an observed supernova is significant because long GRBs associated with massive-star collapse are commonly expected to show supernova emission (Troja et al., 2022; Gompertz et al., 2022). Its absence therefore weakens a straightforward collapsar interpretation, especially when combined with the kilonova-like excess.
- **Host-galaxy and offset properties compatible with a compact binary origin.** The host environment and spatial offset provide population-level context. They do not prove a merger origin on their own, but they are compatible with a compact binary system that may have travelled from its birth site before merging.
- **X-ray and high-energy signatures suggestive of prolonged central-engine activity.** Late X-ray or high-energy activity indicates that the engine may remain active beyond the initial prompt phase. This is directly relevant to the present paper because a long-lived or temporarily stable magnetar could provide one possible explanation for such extended engine behaviour.

Taken together, these properties motivate the question of whether the central engine of GRB 211211A, if magnetar-powered, would resemble the high-field short-GRB magnetar population rather than the ordinary long-GRB population.

4.2 What the B_p–P_0 plane would test

The B_p–P_0 plane would not by itself prove the origin of GRB 211211A. However, it would provide an engine-level consistency test. If plateau or late-injection observables allow an estimate of L_0 and τ, then one may infer B_p and P_0 and compare them with the short- and long-GRB loci. The strength of this test depends on whether a plateau or late-injection component can be robustly isolated from the afterglow contribution (Hamidani et al., 2024; Kunert et al., 2024).

The relevant diagnostic question is:

> Does a long-duration, merger-like GRB occupy the high-field short-GRB magnetar locus, the lower-field long-GRB locus, or an intermediate region?

If GRB 211211A were found to occupy the short-GRB magnetar locus, this would strengthen the interpretation that duration is misleading and that the engine is more closely related to merger-driven systems. If it occupied an intermediate region, the event would become a particularly useful probe of transitional central-engine populations.

4.3 Implication

GRB 211211A motivates the main claim of this paper: anomalous long-duration GRBs should be tested against central-engine parameter spaces, not classified by duration alone.

5 Motivating Case II: GRB 250702B as an Ultra-long, Multi-episode Magnetar Candidate

5.1 Summary of the case

GRB 250702B presents a different kind of anomaly (Girard, 2026b; Granot et al., 2026). Unlike GRB 211211A, whose main difficulty lies in the mismatch between duration and merger-like signatures, GRB 250702B is anomalous because of its extreme temporal structure, obscuration, and host-galaxy context. These properties make it a useful test case for asking whether a single magnetar engine can remain physically plausible under unusually demanding conditions. The event displays:

- **Three bright gamma-ray episodes.** The burst is not characterized by a single continuous prompt phase, but by multiple bright emission episodes. This structure raises the question of whether the engine underwent repeated intrinsic reactivation, or whether a more continuous engine was modulated by fallback, geometry, or viewing conditions.
- **A quasi-regular separation of approximately**

  $$P \simeq 2825 \text{ s.}$$

  The roughly repeated spacing between episodes suggests the presence of a characteristic timescale. This does not by itself prove a periodic engine, but it motivates models involving fallback modulation, disk or jet precession, or other geometrical mechanisms capable of producing quasi-regular visibility windows.
- **A total duration of order several hours.** The total duration places GRB 250702B well outside the ordinary prompt-emission timescale of most GRBs. Any proposed engine must therefore account not only for the energy release, but also for the ability to sustain or modulate activity over an unusually long interval.
- **An extremely obscured afterglow.** Strong obscuration complicates the interpretation of the optical and near-infrared counterpart. It may hide components that would otherwise help distinguish between progenitor scenarios, such as a supernova, kilonova-like excess, or dust-reprocessed emission. This makes the central-engine question more important, because some environmental diagnostics are weakened.

- **An off-nuclear position in a massive, dust-rich, morphologically disturbed host galaxy.** The host context suggests that the event occurred in a complex and possibly non-axisymmetric environment. This does not determine the progenitor or engine by itself, but it makes environmental modulation, structured fallback, and geometrical effects more plausible ingredients in the interpretation.

A proposed interpretation invokes a single newborn magnetar embedded in a non-axisymmetric, merger-driven environment. In this picture, clumpy fallback accretion provides intermittent energy release, while slow precession of the jet or fallback disk produces quasi-periodic visibility modulation rather than requiring repeated intrinsic engine shutdowns and restarts (Girard, 2026b).

5.2 What the B_p–P_0 plane can and cannot test

The B_p–P_0 plane does not explain the quasi-periodicity by itself. It does not test environmental torques, disk precession, or host-induced asymmetry directly. Thus, the B_p–P_0 plane constrains the plausibility of the proposed engine, not the geometrical origin of the observed episode spacing. What it can test is whether the proposed single-magnetar engine occupies a plausible region of magnetar parameter space. For GRB 250702B, the relevant question is:

> What B_p and P_0 would be required for a single magnetar to power such an ultra-long, multi-episode event?

Several outcomes are possible:

- **Parameters near the long-GRB locus.** If the inferred parameters lie near the long-GRB locus, the stripped-star magnetar scenario remains naturally aligned with collapsar-like engines. In this case, GRB 250702B would be anomalous mainly because of its duration, episode structure, obscuration, and host environment, rather than because it requires a fundamentally different magnetar engine.

- **Parameters approaching the high-field short-GRB locus.** If the inferred parameters approach the high-field short-GRB locus, the event may require a more unusual engine or formation pathway. Such a result would not automatically imply a compact-merger origin, but it would suggest that the magnetar is not simply an ordinary long-GRB engine stretched over a longer timescale.

- **Extreme outlier parameters.** If the inferred parameters are extreme outliers, the apparent parsimony of the single-magnetar model is weakened (Granot et al., 2026). The model would then require additional assumptions, such as unusually strong magnetic fields, very rapid initial spin, non-standard neutron-star properties, or a more complex engine geometry.

The diagnostic therefore does not decide whether the quasi-regular episode spacing is caused by fallback, precession, or environmental modulation. It asks a more limited but crucial question: whether the magnetar required by the scenario remains physically plausible in the emerging B_p–P_0 engine parameter space.

5.3 Implication

GRB 250702B is not already solved by the B_p–P_0 plane. Rather, it shows why such a plane is useful: it turns a qualitative magnetar proposal into a testable engine-parameter claim.

6 Motivating Case III: GRB 241025A and the Problem of Model Degeneracy

6.1 Why include GRB 241025A?

GRB 241025A is not used here primarily as a magnetar-engine case study. Its role is methodological. The event illustrates how a single observational pattern can support multiple coherent interpretations.

In one interpretation, the radio spectral-index evolution is explained by the passage of synchrotron break frequencies through the radio band in a standard external-shock afterglow (Sari et al., 1998; Giarratana et al., 2026). In another, a Binary-driven HyperNova narrative embeds the same multi-wavelength behaviour within a richer progenitor and multi-component emission scenario (Ruffini et al., 2024).

6.2 Connection to the present argument

This case shows that anomalous GRBs often do not suffer from a lack of models. They suffer from an excess of coherent but under-constrained mappings between data and theory (Girard, 2026c; Girard, 2026d; Girard and Kline, 2026). A standard synchrotron external-shock interpretation and a more complex progenitor-driven narrative may both remain compatible with the same multi-wavelength dataset when the available observables do not uniquely select one physical mapping. This is directly relevant to the present argument. The goal of the B_p–P_0 diagnostic is not to add another narrative to an already crowded interpretive space. Its purpose is to introduce an additional constraint at the level of the central engine. When a magnetar-powered interpretation is proposed, the relevant question becomes whether the required engine parameters occupy a plausible and population-consistent region of magnetar parameter space. GRB 241025A therefore illustrates the broader diagnostic problem: phenomenological adequacy does not necessarily imply engine-level specificity. A model may reproduce the observed light curves or spectral evolution while leaving the nature of the central engine underdetermined. This is precisely the kind of situation in which additional constraints become valuable.

6.3 Limitation

GRB 241025A should not be forced into the same magnetar framework unless the necessary plateau or late-injection observables justify such a test. Its inclusion here is limited to the broader lesson: diagnostics are valuable when they reduce model non-specificity. GRB 241025A is therefore included not as evidence for a magnetar engine, but as a reminder that phenomenological adequacy does not guarantee engine-level specificity. The event functions as a methodological control case: it shows why a diagnostic such as the B_p–P_0 plane should be applied selectively, only when the relevant engine-sensitive observables are present. This limitation is important. If the B_p–P_0 plane were applied indiscriminately, it would risk becoming another weak analogy rather than a specificity-enhancing test. Its value depends on restraint: the diagnostic should be used when the data support a magnetar spin-down interpretation, and withheld when they do not.

The complementary roles of the three motivating cases are summarized in Figure 2.

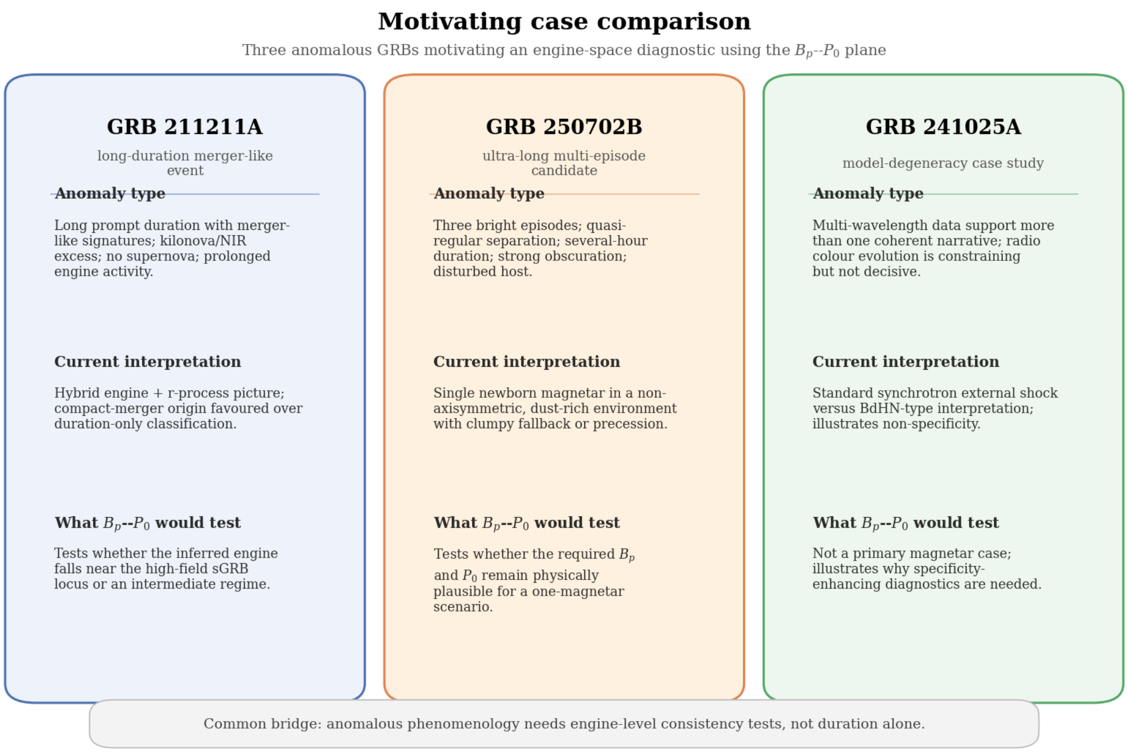

Figure 2: Motivating case comparison. A three-column schematic comparing GRB 211211A, GRB 250702B, and GRB 241025A according to anomaly type, current interpretation, and what the B_p–P_0 diagnostic would test. These three cases do not define a new class of GRBs; they define a diagnostic need. The common problem is not anomaly itself, but the absence of an engine-level constraint.

7 Proposed Diagnostic Workflow

7.1 When to apply the test

The B_p–P_0 diagnostic is most relevant when an event exhibits one or more observables that can plausibly be linked to prolonged central-engine activity. The test should not be applied automatically to every anomalous GRB. It becomes useful when the data contain a feature from which a magnetar spin-down luminosity, timescale, or engine-level constraint may reasonably be inferred.

- **An internal X-ray plateau.** An internal X-ray plateau is the clearest case in which the B_p–P_0 diagnostic may be applied (Lyons et al., 2010; Rowlinson et al., 2014; Lü et al., 2018; Li et al., 2026). If the plateau can be associated with magnetar spin-down rather than with the external afterglow, its luminosity and break time can provide the basic observables needed to infer L_0, τ, B_p, and P_0.

- **A steep decay after the plateau.** A steep post-plateau decline strengthens the case for an internal origin, because it may indicate the abrupt fading or collapse of the central engine. This makes the plateau more useful as an engine diagnostic, although the interpretation still depends on separating internal activity from afterglow evolution.

- **Extended emission.** Extended emission indicates that the burst activity continues beyond the initial prompt phase. It may point to a long-lived engine, such as a temporarily stable magnetar or fallback-powered system. However, extended emission alone does not identify the engine; it only motivates a more detailed engine-parameter test.

- **Late-time energy injection.** Late energy injection can flatten or re-energize the afterglow and may indicate continued power from the central engine. If the injection history can be modelled, it may provide indirect constraints on the duration and luminosity of the engine activity relevant to the B_p–P_0 plane.

- **X-ray or radio flattening.** Flattening in X-ray or radio light curves may signal refreshed shocks, density structure, viewing-angle effects, or continued engine activity. Because several explanations are possible, this feature is not diagnostic by itself. It becomes relevant when the flattening is consistent with a coherent late-injection or magnetar-powered scenario.

- **Rebrightening plausibly linked to a central engine.** Rebrightening episodes can indicate renewed energy release or delayed interaction between ejecta and the external medium. If the timing and spectral behaviour favour an engine origin, such episodes may help constrain whether the required magnetar parameters remain plausible.

- **Multi-wavelength evidence for prolonged engine activity.** The strongest application occurs when several bands point toward the same conclusion: continued activity beyond the prompt phase. Consistency across X-ray, optical, near-infrared, radio, or high-energy observations reduces the risk of over-interpreting a single-band feature and strengthens the case for applying the B_p–P_0 diagnostic.

In this sense, the B_p–P_0 test should be applied selectively. Its role is not to classify every anomalous GRB, but to evaluate whether a proposed magnetar-powered interpretation remains physically plausible when engine-sensitive observables are available.

The proposed workflow is summarized schematically in Figure 3.

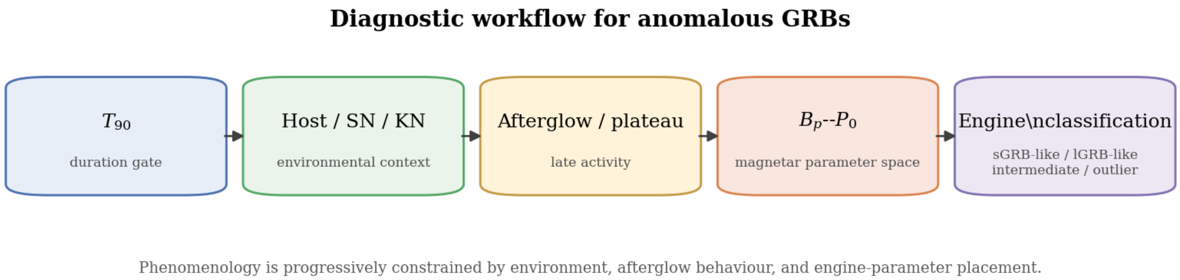

Figure 3: Diagnostic workflow for anomalous GRBs. The interpretation proceeds from duration-based classification through environmental and afterglow diagnostics, then toward magnetar-engine parameter placement in the B_p–P_0 plane and final engine classification.

7.2 Operational steps

The proposed workflow is as follows:

1. **Identify engine-sensitive observables.**
   Determine whether the event contains an X-ray plateau, a break time, late injection, or other signatures plausibly linked to central-engine activity.

2. **Estimate the plateau luminosity and spin-down timescale.**
   If a plateau is present, estimate
   $$L_0 \simeq L_b$$
   and
   $$\tau = \frac{t_b}{1+z}.$$
   (Li et al., 2026)

3. **Infer magnetar parameters** (Lyons et al., 2010; Rowlinson et al., 2014; Li et al., 2026)**.**
   Use the spin-down relations to derive B_p and P_0.

4. **Place the event in the B_p–P_0 plane.**
   Compare the inferred point or allowed region to the short-GRB and long-GRB loci.

5. **Interpret the locus.**
   Classify the engine as short-GRB-like, long-GRB-like, intermediate, or outlying in parameter space.

6. **Cross-check with independent diagnostics.**
   Compare the engine inference with host properties, supernova or kilonova evidence, afterglow behaviour, radio evolution, polarization, and late high-energy activity.

7.3 What the test does not do

The B_p–P_0 plane should not be treated as a complete progenitor classifier. It does not by itself prove a merger origin, a collapsar origin, a black-hole engine, a magnetar engine, jet precession, or environmental modulation. It tests whether a magnetar interpretation, when invoked, occupies a plausible and informative region of engine parameter space.

8 Discussion

8.1 A bridge between three problems

The proposed diagnostic links three related problems. These problems are not identical, but they converge on the same methodological need: anomalous GRBs require constraints that connect phenomenology to central-engine physics.

- **GRB 211211A shows that a long-duration burst can exhibit merger-like properties** (Troja et al., 2022; Gompertz et al., 2022; Hamidani et al., 2024; Girard, 2026a)**.** This case demonstrates that T_{90} is not always a reliable proxy for progenitor channel. A burst may appear long by duration while displaying kilonova-like, host, offset, or prompt-emission properties more naturally associated with compact-object mergers. GRB 211211A therefore motivates the need for an engine-level test that can ask whether a long-duration event is nevertheless compatible with the short-GRB magnetar population.

- **GRB 250702B shows that ultra-long, multi-episode events may require unusual central-engine explanations** (Girard, 2026b; Granot et al., 2026)**.** This case pushes the problem in a different direction. Its extreme duration, multi-episode structure, obscuration, and disturbed host environment make a simple phenomenological classification inadequate. The B_p–P_0 plane does not explain the episode spacing, but it can test whether the proposed single-magnetar engine remains physically plausible under such demanding conditions.

- **GRB 241025A shows that distinct theoretical narratives can remain compatible with the same multi-wavelength dataset** (Giarratana et al., 2026; Ruffini et al., 2024; Girard and Kline, 2026)**.** This case illustrates the problem of model non-specificity. A dataset may be observationally rich while still allowing more than one coherent interpretation. GRB 241025A is therefore not included as a primary magnetar case, but as a reminder that phenomenological adequacy does not guarantee engine-level specificity.

The B_p–P_0 plane provides a way to translate these classification problems into an engine-space question. It does not eliminate the need for duration, host, kilonova, supernova, or afterglow diagnostics. Rather, it asks whether a proposed magnetar interpretation occupies a physically plausible region of parameter space relative to known short- and long-GRB engine populations.

8.2 Relation to black-hole and fallback models

A magnetar-compatible location in the B_p–P_0 plane does not exclude black-hole or fallback scenarios. Conversely, a poor fit to the magnetar parameter space does not automatically validate a black-hole model. The diagnostic should be used comparatively. It asks whether the magnetar version of a given interpretation is physically plausible and population-consistent. This distinction is important because "black hole engine" can otherwise become a residual explanation: a broad label invoked whenever a magnetar interpretation appears difficult, without specifying which observable requires a black hole, which fallback history is implied, or which predictions distinguish the model from a magnetar-powered alternative. In that form, the black-hole interpretation risks functioning as an explanatory catch-all rather than as a discriminating physical model. The B_p–P_0 plane therefore does not function as a winner-takes-all classifier. Its role is narrower but useful: it tests whether the magnetar hypothesis remains viable when translated into engine parameters. If an event falls within a plausible magnetar locus, alternative engines are not ruled out; the result only shows that the magnetar interpretation survives one important consistency check. If an event falls outside the plausible magnetar region, this does not prove a black-hole engine; it shows that the magnetar interpretation requires additional assumptions or should be compared more seriously with fallback or black-hole alternatives. Thus, the proposed diagnostic also imposes symmetry on the comparison. A magnetar model should not be accepted merely because it is elegant, but a black-hole or fallback model should not be accepted merely because it is flexible (Granot et al., 2026; Hamidani et al., 2024). Each engine hypothesis should be required to specify the observables it explains, the parameter regime it requires, and the predictions by which it could be distinguished from competing scenarios.

8.3 Limitations

Several limitations must be emphasized:

- **B_p and P_0 are model-dependent inferred quantities.** These parameters are not measured directly. They are derived from plateau luminosity, timescale, redshift, and assumptions about magnetar spin-down. The resulting position in the B_p–P_0 plane is therefore only as reliable as the underlying model and observational inputs.

- **Not all X-ray plateaus are necessarily internal or magnetar-powered.** Some plateaus may arise from external afterglow physics, viewing-angle effects, energy injection into the blast wave, or environmental structure. The B_p–P_0 diagnostic is strongest only when the plateau can be plausibly associated with internal central-engine activity.

- **Uncertain redshifts directly affect luminosity and timescale estimates.** The inferred plateau luminosity and rest-frame spin-down timescale depend on the redshift. When the redshift is unknown or uncertain, the derived magnetar parameters can shift substantially, weakening the diagnostic value of the placement.

- **Canonical neutron-star assumptions influence the inferred values.** The derivation of B_p and P_0 usually assumes fiducial values for neutron-star radius, moment of inertia, and radiative efficiency. Different assumptions can move the inferred parameters in the plane, especially for borderline or outlier cases.

- **A correlation in an inferred parameter plane may partly reflect the structure of the model.** Because both B_p and P_0 are derived from related observables, an apparent correlation may partly encode the mathematical structure of the spin-down model. The B_p–P_0 plane should therefore be treated as a diagnostic framework, not as a model-independent empirical boundary.
- **The method is strongest when applied to events with clear plateau or late-injection observables.** The diagnostic loses force when the relevant light-curve features are weak, poorly sampled, contaminated by afterglow emission, or absent. In such cases, the B_p–P_0 plane may still guide interpretation, but it should not be overused.

These limitations do not undermine the diagnostic. They define its proper domain of application. The B_p–P_0 plane is most useful when treated as a conditional consistency test: if a magnetar engine is invoked, then the inferred parameters should be physically plausible and comparable to known GRB engine populations.

8.4 Observational priorities

Future anomalous GRBs should be followed with the goal of constraining both phenomenology and engine parameters. Particularly valuable observations include:

- **Rapid X-ray follow-up to detect early plateaus and break times** (Lyons et al., 2010; Rowlinson et al., 2014; Li et al., 2026)**.** Early X-ray coverage is essential for identifying internal plateaus and measuring their break times. Without this information, the spin-down timescale τ cannot be robustly estimated, and the B_p–P_0 diagnostic becomes much weaker.
- **Prolonged X-ray monitoring to measure post-plateau decay.** Continued X-ray monitoring helps determine whether the plateau is followed by a steep decline, a gradual afterglow-like decay, or more complex late activity. This distinction matters because a steep post-plateau decay can strengthen the case for an internal engine origin.
- **Early optical and ultraviolet observations to detect blue kilonova or engine-heated components.** Early optical/UV observations can reveal fast, blue components that may be associated with engine heating or merger ejecta. Such detections help connect the engine interpretation to the ejecta and progenitor environment.
- **Deep near-infrared monitoring to constrain red kilonova emission** (Troja et al., 2022; Gompertz et al., 2022; Hamidani et al., 2024; Waxman et al., 2025)**.** Near-infrared follow-up is crucial for detecting or excluding red kilonova components. This is particularly important for long-duration merger-like events, where a kilonova signature can challenge a simple collapsar interpretation.
- **Late radio and millimetre monitoring to detect flattening, rebrightening, or deviations from standard afterglow decay** (Sari et al., 1998; Giarratana et al., 2026)**.** Radio and millimetre observations probe late energy injection, refreshed shocks, density structure, and possible long-lived engine activity. These data can help determine whether a magnetar-powered interpretation remains necessary or whether standard afterglow evolution is sufficient.

- **Polarization measurements to constrain geometry and magnetic-field structure.** Polarization can provide information about jet geometry, magnetic-field ordering, and emission-region structure. While it does not directly measure B_p or P_0, it can help distinguish between engine-driven, geometric, and environmental interpretations.

- **Host-galaxy characterization, including morphology, offset, dust content, and star-formation environment.** Host observations provide the environmental context needed to interpret the burst. Morphology, offset, dust content, and star-formation rate can strengthen or weaken merger-like, collapsar-like, or disturbed-environment scenarios, and should be compared with the inferred engine parameters.

The observational goal is not simply to collect more data, but to collect data that reduce ambiguity. The most valuable follow-up observations are those that connect the phenomenology of an anomalous GRB to a testable central-engine interpretation.

9 Conclusion

The short/long GRB classification based on T_{90} remains useful but is no longer sufficient for anomalous events that combine duration, host, kilonova, supernova, and afterglow signatures in non-canonical ways. The emerging B_p–P_0 parameter space of magnetar-powered GRBs offers a complementary diagnostic axis.

This paper proposes that anomalous GRBs with plausible magnetar-powered activity should be evaluated by their inferred or expected location in the B_p–P_0 plane. Such a test does not replace traditional diagnostics. Rather, it adds a central-engine layer to them.

The main conclusion may be stated as follows:

> The B_p–P_0 plane should be treated as a third diagnostic axis, alongside duration and environment, for anomalous GRBs in which magnetar-powered activity is observationally plausible.

Future anomalous GRBs should therefore be evaluated not only by what they look like phenomenologically, but also by where their engines fall in the emerging parameter space of magnetar-powered transients.

Tables

Diagnostic	**Observable**	**Strength**	**Failure mode**
T_{90}	Prompt duration	Simple first classification	Fails for long merger-like GRBs
Kilonova / supernova	Optical/NIR transient components	Strong progenitor clue	Can be faint, obscured, or missed
Host / offset	Environment and location	Useful population context	Indirect, not decisive alone
X-ray plateau	Late engine activity	Directly tied to engine models	Model-dependent interpretation
B_p–P_0	Magnetar engine parameters	Quantitative engine-space test	Requires suitable plateau or injection data

Table 1: Diagnostic axes for anomalous GRBs and their main limitations.

Event	**Anomaly**	**Current interpretation**	B_p–P_0 **test**
GRB 211211A	Long duration with merger-like signatures	Hybrid engine plus kilonova/r-process picture	Test whether the engine is short-GRB-like or intermediate
GRB 250702B	Ultra-long, multi-episode, obscured event	Single magnetar in disturbed environment	Test whether the required magnetar is physically plausible
GRB 241025A	Model degeneracy in multi-wavelength interpretation	Synchrotron external shock versus BdHN-type narrative	Illustrates need for specificity-enhancing diagnostics

Table 2: Motivating cases and the role of the B_p–P_0 diagnostic.

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